

SAI KARTHIK K

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github.com/Saikarthik63 ♦ sai-karthik-k-portfolio.vercel.app

SUMMARY

M.Tech student specializing in Embedded Systems Technology with a strong foundation in embedded systems, firmware development, software development, Internet of Things (IoT), and Artificial Intelligence (AI). Proficient in C, C++, Embedded C, Python, and SQL, with hands-on experience in RTOS, Embedded Linux, hardware–software integration, computer vision, sensor interfacing, and automotive embedded systems. Passionate about designing scalable, reliable, and intelligent technology solutions, and seeking opportunities in Embedded Systems, Firmware Engineering, Software Development, AI/ML, IoT, and Automotive Embedded Systems.

EDUCATION

SRMIST - Kattankulathur 2027

M.Tech, Embedded System Technology · CGPA - 8.46/10

St. Joseph's College of Engineering 2025

B.E, Electronics and Communication Engineering (ECE) · Percentage - 68.8%

New Prince Matriculation Hr. Sec. School 2021

Class XII, Tamil Nadu State Board · Percentage - 80.2%

New Prince Matriculation Hr. Sec. School 2019

Class X, Tamil Nadu State Board · Percentage - 70%

SKILLS

Programming Languages: C, C++, Embedded C, Python, SQL, Core Java

Tools and Technologies: Excel, PowerPoint, MATLAB, Keil uVision, STM32CubeIDE, MPLAB X IDE, Xilinx Vivado, Proteus, EasyEDA, Visual Studio Code, Jupyter Notebook, Google Colab, Linux, MySQL, Power BI,

Domain Knowledge: Embedded Systems, Internet of Things (IoT), Automotive Embedded Systems, Computer Science Fundamentals, Software Development, Artificial Intelligence (AI), Machine Learning (ML)

Technical Skills: Firmware Development, Software Development, Embedded Linux, RTOS, Hardware-Software Integration, Sensor Interfacing, Communication Protocols, Debugging and Testing, Hardware Debugging, Computer Vision, Problem Solving.

EXPERIENCE

PCB Design & Embedded Hardware Intern Jan 2025 - Mar 2025

Mirror Technologies Pvt. Ltd.

Ashok Nagar, Chennai

- Assisted in PCB design, schematic analysis, component placement, and layout verification.
- Supported embedded hardware development through prototype assembly, testing, and hardware validation.
- Performed circuit debugging and troubleshooting to ensure reliable hardware performance.

Embedded Systems Intern Jun 2024 - Jun 2024

EMCOG Solutions Pvt. Limited

Avadi, Chennai

- Assisted in developing Embedded C firmware and hardware–software integrated applications.
- Integrated sensors and peripherals for real-time embedded system development.
- Performed debugging, testing, and validation to ensure reliable system performance.

PROJECTS

Hand Gesture-Based Volume Control System Feb 2024 - Apr 2024

Computer Vision & AI · Tools: Python, OpenCV, MediaPipe

- Developed a real-time hand gesture-based volume control system using Python, OpenCV, and MediaPipe.
- Implemented hand tracking and gesture recognition for touch-free volume adjustment.

AI Voice Assistant Using ESP32

Aug 2024 - Apr 2025

IoT & AI · Tools: ESP32, Embedded C, Voice Recognition, Wireless Communication

- Developed an ESP32-based AI voice assistant for real-time voice command recognition and device control.
- Integrated voice recognition and wireless communication for smart IoT device automation.

Smart Indoor Environment Monitoring System

Oct 2025 - Nov 2025

Embedded Systems & IoT · Tools: ESP32, Embedded C, Sensor Interfacing, IoT

- Developed an ESP32-based multi-sensor indoor environment monitoring system.
- Implemented real-time environmental monitoring with an Indoor Health Index (IHI) and automated alerts.

Energy-Aware Adaptive Irrigation Control Using Edge Computing

Mar 2026 - Apr 2026

Embedded Systems & IoT · Tools: ESP32, Embedded C, Edge Computing, Soil Moisture Sensor

- Developed an ESP32-based smart irrigation system with adaptive irrigation control.
- Implemented real-time soil moisture monitoring and energy-efficient irrigation using ESP32 deep sleep mode.

CERTIFICATIONS

VLSI Design | [CERTIFICATE](#)

Jun 2023

Udemy

Embedded Systems | [CERTIFICATE](#)

Nov 2025

LaunchED Global · LEDCCN1280

Foundations of Prompt Engineering

Jun 2026

Amazon Web Services (AWS)

Claude 101 | [CERTIFICATE](#)

Jun 2026

Anthropic

Introduction to Embedded Linux | [CERTIFICATE](#)

Jun 2026

Microchip University

Academy Accreditation - Generative AI Fundamentals | [CERTIFICATE](#)

Jul 2026

Databricks · 188170754

LANGUAGES

Tamil [Native Proficiency], English [Professional Working Proficiency]